

Programming Touch as a Somatic Activity

Yoav Luf

▶ **To cite this version:**

Yoav Luf. Programming Touch as a Somatic Activity. Doctoral Consortium at the sixth decennial Aarhus conference: Computing X Crisis, Aug 2025, Aarhus, Denmark. ⟨hal-05191120⟩

HAL Id: hal-05191120

https://hal.science/hal-05191120v1

Submitted on 29 Jul 2025

HAL is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.



HAL Authorization

Programming Touch as a Somatic Activity

Yoav Luft
luft@kth.se
Royal Institute of Technology KTH
Stockholm, Sweden

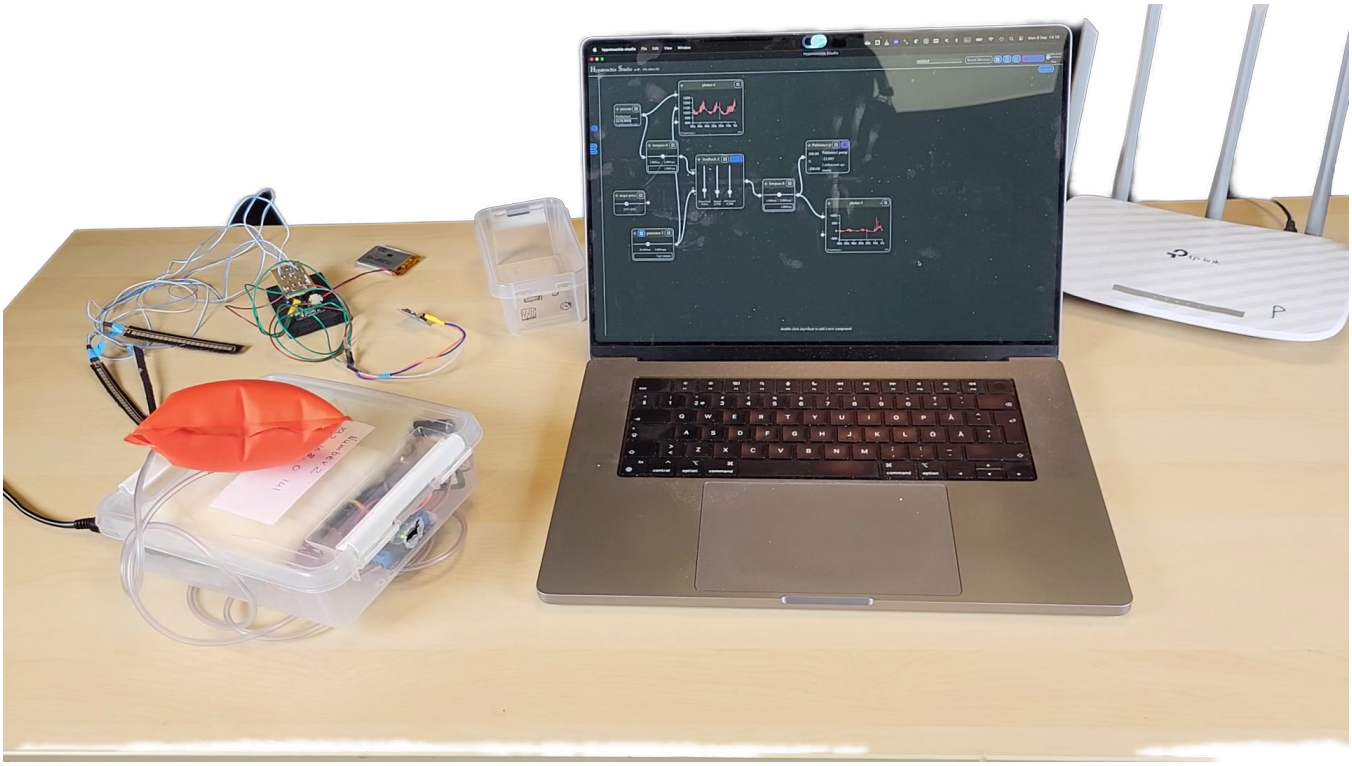


Figure 1: Hypatouchia Studio set up to work with a pneumatic bit.

Abstract

As technology moves closer and closer to our bodies, the need to design refined, thoughtful and rich experiences grows. My project focuses on an aspect of designing such interactions that is the shaping of their behaviour, often in the form of computer programmes that integrate sensors and other data sources with actuator controllers. However, programming interactive touch experiences has proved to be a difficult task, requiring the integration of designerly knowledge of the experiences being shaped and technical expertise in choosing and implementing algorithmic solutions to achieve the desired experience.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
Aarhus '25 Doctoral Consortium, Aarhus, Denmark
© 2025 Copyright held by the owner/author(s).

CCS Concepts

• **Human-centered computing → Systems and tools for interaction design**; *Empirical studies in interaction design*; *Interaction design process and methods.*

Keywords

Soma design, Programming, Interaction design, Computational composites

ACM Reference Format:
Yoav Luft. 2025. Programming Touch as a Somatic Activity. In *Proceedings of Aarhus '25 Doctoral Consortium*. ACM, New York, NY, USA, 4 pages.

1 Introduction

As technology moves closer and closer to our bodies, the need to design refined, thoughtful and rich experiences grows. My project focuses on an aspect of designing such interactions that is the shaping of their behaviour, often in the form of computer programmes that integrate sensors and other data sources with actuator controllers. However, programming interactive touch experiences has

proved to be a difficult task,[10, 15] requiring the integration of designerly knowledge of the experiences being shaped and technical expertise in choosing and implementing algorithmic solutions to achieve the desired experience [20].

Motivated by this challenge, I engage with a three-part research question: (a) how can computational materials be explored as a design material for digital touch? (b) How to support somatic sensibilities in the design of computations for touch by technology? And (c) what qualities of the computational material can be shaped by programming it? These questions resonate with similar inquiries, for example, inquiries into the shaping of digital materials [21] using imagined tools, the question of programming for moving bodies [6] and questions of bringing forth feminist values into the field of programming [8].

Part of the ERC project for Designing Intimate Touch, I tackle these questions in several studies to explore and understand how the computational materials involved in designing touch-on-the-body from technology can be explored richly and productively. As the details of material exploration, and specifically exploration of computational material, is often omitted from publications, I interviewed design researchers on their experiences exploring such material. To be able to observe how exploration is enacted in-situ, I designed Hypatouchia Studio, a curated visual programming studio compatible, or supporting easy compatibility, with other toolkits such as Soma Bits [24], Tactorbots [25], Emotibit [17], and any other Arduino based toolkit.

1.1 Related Works

These is a body of work addressing different aspects of designing behaviours of technological artefacts, some of which intersects with touch, for example in the field of haptics, while other works address designers directly or as part of a category of programmers who work outside of software engineering practices.

Currently, the design of touch from technology behaviours is addressed by designing pre-recorded movements, as demonstrated, for example, by CoTacs [15] and Tactorbots [25] using rich authoring tools. Projects like Tactile Symbols [18] or Feelix [22] address designing for feedback control with specific technologies such as vibration motors and stepper motors. Other projects try to address other aspects of programming through storytelling [16] or state-machines [14] without focusing on touch specifically, to name a couple.

A tangent field to learn from is that of end-user programming, end-user software development and end-user software engineering [13]. This field of research, closely tied to research in programming languages and psychology of programming,[5] in that it tries to understand how people conceptualise, design, debug, and use software. End-user programming had addressed both visual programming,[3, 7] the paradigm used in my research, as well as exploring the role of AI in design [4].

2 Project Goals

I intended to probe into my research questions with a research through design approach, producing new tools to support designers in the exploration of computational materials. I hope to produce both tools and methodologies to cultivate designer-engineer collaboration for shaping the behaviours of computational materials, to cultivate tacit somatic understanding of how a certain algorithm might affect the interpretation of signals from the body, and the reproduction of data in the form of algorithmic control of computational materials on the body.

By enriching our methodologies and improving our tools, I hope to allow future designers to design touch experiences that exhibit thoughtful, rich, dignified, articulate and novel behaviours of touch from technology.

In this ecosystem of tools, I identify the need to support the early exploration of computational materials – sensors, actuators and algorithms – to be done early in the project. This is to allow designers to consider what materials are most appropriate for their design goals without over-committing to certain technologies or approaches, and to bring to the front of the exploration embodied, somatic and ethical knowledge of the design team. In my vision, the behaviours and qualities of computational materials can be quickly explored by coupling them to one another, applying different algorithmic techniques, adjusting parameters, and doing all of this in situ while interacting with the materials without being forced to think of the implementation details or specific calculations. I model this exploration of computational material on the ideas presented in soma design [9] and the idea of sketching with computational materials [23].

2.1 Contributions

A central contribution of my research is towards discovering ways to centre our soma in the design of computational materials behaviour. Current approaches to programming, including those focused on end-user programming, sideline the body and focus on programming – the act of designing interactions – as a linguistic, cognitivist activity. Through exploring different modes of programming and developing a tacit understanding of algorithms, I hope to close the gap between interacting bodily with a computational system and shaping it.

Centring on the body in the act of programming also relates to feminist discourse. By thinking of the body as intimately involved in programming systems, we open up the stage to questions of identity of the programmer-designer as a gendered, racialised, physical and interdependent being. My goal is to rethink programming otherwise, not as a disembodied cognitive task of expressing knowledge and behaviour through mathematically pure models and logic, but rather as a lived, evolving capture of our lived experiences as social somatic beings.

3 State of the Project

I have completed two studies, which are pending publication. For the study, I interviewed 13 design-researcher individuals or teams about their practices of exploring and sense-making of computational material. This study reveal both an expected diversity of approaches, but also surprising communalities, such as the centrality of first-person knowledge that runs across the different epistemologies of the interviewees. My second study involved the design of Hypatouchia Studio, a visual programming environment with a curated set of building blocks drawn from the previous study and

my own personal experience as a research engineer. Hypatouchia Studio was used in the context of two workshops: the first an open-ended explorative design workshop and the second workshop aimed at sensor exploration for an independent design research. These workshops were conducted in order to better understand the needs of programming in the context of computational materials in soma design.

3.1 Future Work

Planned research is focused on long-term, first-person exploration of programming touch-on-the-body from technology.

For the remainder of my PhD, I intend to explore some of the following, presented with relevant methods:

- Exploring the integration of tangible elements into programming environments using research through design and soma design.
- Exploring the first-person experiences of programming touch on the body from technology using longitudinal diary studies, interviews, autobiographical methods and possibly post-phenomenological methodology (through collaboration with appropriate experts). I already conducted a pilot of such a study.
- Do critical design [2] with different machine learning techniques in the context of programming for touch. This depends on being able to collaborate with experts.

I would like to use this doctoral consortium to process and think of my plans for the second part of PhD programme.

3.2 Computing Programming Crisis

One particular emerging issue is the proliferation of large language model systems (LLMS) into all aspects of creative work. While these systems do present impressive and continuously evolving capabilities to generate structurally cohesive text, including computer programmes and visuals based on textual analysis, they raise moral and practical concerns [4]. Concerns address both whose knowledge and expertise these systems claim to replace, what and whose knowledge and expertise they were trained and therefore what kind of knowledge is embedded in them. Additionally, the training and execution of LLMs is more resource demanding, therefore presenting an ecological concern as well.

I hope to challenge the emerging dominance of LLMs in programming by centring on the rarely discussed embodied nature of programming as design work. This embodied nature is especially important when addressing the design of systems engaged with touch, as touch is such an intimate and discrete sense [15] and difficult to articulate verbally [19]. In my studies, I observed how embodied action plays an important role in communicating and sense-making of techno-social systems, often in a non-verbal way. My observations are in line with earlier observations on the enacted aspects of programming [11].

An open question there is on the role of the current trends of machine learning in designing such systems. Can machine learning supported programming help or hinder the design of touch by technology? While some projects tried to incorporate machine learning with soma design approaches [1, 12] this approach had not yet bear fruits, to the best of my knowledge. Inspired by the critique of Alan Blackwell, a pioneer in the field of psychology of programming, I would like to explore how we can programme better and how can the systems we programme can be more nuanced, insightful and encompass our lived and somatic experiences.

Acknowledgments

This work was partially supported by the European Union (ERC, Intimate Touch, 101043637). Views and opinions expressed are however those of the author only and do not necessarily reflect those of the European Union or the European Research Council.

References

[1] Miquel Alfaras, Vasiliki Tsaknaki, Pedro Sanches, Charles Windlin, Muhammad Umair, Corina Sas, and Kristina Höök. 2020. From Biodata to Somadata. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)*. Association for Computing Machinery, New York, NY, USA, 1–14. doi:10.1145/3313831.3376684

[2] Jeffrey Bardzell and Shaowen Bardzell. 2013. What is "critical" about critical design?. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '13)*. Association for Computing Machinery, New York, NY, USA, 3297–3306. doi:10.1145/2470654.2466451

[3] Andrea Bellucci, Paloma Díaz, and Ignacio Aedo. 2023. Exploring Visual Languages for Prototyping Interactive Behaviors for Tangible Virtual Reality. In *End-User Development (Lecture Notes in Computer Science)*, Lucio Davide Spano, Albrecht Schmidt, Carmen Santoro, and Simone Stumpf (Eds.). Springer Nature Switzerland, Cham, 203–219. doi:10.1007/978-3-031-34433-6_13

[4] Alan F. Blackwell. 2024. *Moral codes: designing alternatives to AI.* The MIT Press, Cambridge.

[5] Alan F. Blackwell, Marian Petre, and Luke Church. 2019. Fifty years of the psychology of programming. *International Journal of Human-Computer Studies* 131 (Nov. 2019), 52–63. doi:10.1016/j.ijhcs.2019.06.009

[6] Cristian Bogdan, Vasiliki Tsaknaki, Charles Windlin, Marianela Ciolfi Felice, Ozgun Kilic Afsar, Sara Eriksson, Ylva Fernaeus, and Pedro Sanches. 2020. Programming for Moving Bodies. In *Proceedings of the 11th Nordic Conference on Human-Computer Interaction: Shaping Experiences, Shaping Society (NordiCHI '20)*. Association for Computing Machinery, New York, NY, USA, 1–3. doi:10.1145/3419249.3420069

[7] Enrique Coronado, Fulvio Mastrogiovanni, Bipin Indurkhya, and Gentiane Venture. 2020. Visual Programming Environments for End-User Development of intelligent and social robots, a systematic review. *Journal of Computer Languages* 58 (June 2020), 100970. doi:10.1016/j.cola.2020.100970

[8] Felienne Hermans and Ari Schlesinger. 2024. A Case for Feminism in Programming Language Design. In *Proceedings of the 2024 ACM SIGPLAN International Symposium on New Ideas, New Paradigms, and Reflections on Programming and Software*. ACM, Pasadena CA USA, 205–222. doi:10.1145/3689492.3689809

[9] Kristina Höök. 2018. *Designing with the Body: Somaesthetic Interaction Design.* The MIT Press, Cambridge, Massachusetts.

[10] Carey Jewitt, Sara Price, Jürgen Steimle, Gijs Huisman, Lili Golmohammadi, Narges Pourjafarian, William Frier, Thomas Howard, Sima Ipakchian Askari, Michela Ornati, Sabrina Panëels, and Judith Weda. 2021. Manifesto for Digital Social Touch in Crisis. *Frontiers in Computer Science* 3 (Oct. 2021). doi:10.3389/fcomp.2021.754050 Publisher: Frontiers.

[11] Martin Jonsson, Jakob Tholander, and Ylva Fernaeus. 2009. Setting the stage – Embodied and spatial dimensions in emerging programming practices. *Interacting with Computers* 21, 1 (Jan. 2009), 117–124. doi:10.1016/j.intcom.2008.10.004

[12] Ozgun Kilic Afsar, Yoav Luft, Kelsey Cotton, Ekaterina R. Stepanova, Claudia Núñez-Pacheco, Rebecca Kleinberger, Fehmi Ben Abdesslem, Hiroshi Ishii, and Kristina Höök. 2023. Corsetto: A Kinesthetic Garment for Designing, Composing for, and Experiencing an Intersubjective Haptic Voice. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. ACM, Hamburg Germany, 1–23. doi:10.1145/3544548.3581294

[13] Amy J. Ko, Robin Abraham, Laura Beckwith, Alan Blackwell, Margaret Burnett, Martin Erwig, Chris Scaffidi, Joseph Lawrance, Henry Lieberman, Brad Myers, Mary Beth Rosson, Gregg Rothermel, Mary Shaw, and Susan Wiedenbeck. 2011. The state of the art in end-user software engineering. *ACM Comput. Surv.* 43, 3 (April 2011), 21:1–21:44. doi:10.1145/1922649.1922658

[14] Germán Leiva, Nolwenn Maudet, Wendy Mackay, and Michel Beaudouin-Lafon. 2019. Enact. *ACM Transactions on Computer-Human Interaction (TOCHI)* (May 2019). doi:10.1145/3310276 Publisher: ACM PUB27 New York, NY, USA.

[15] Moritz Alexander Messerschmidt, Juan Pablo Forero Cortes, and Suranga Nanayakkara. 2024. CoTacs: A Haptic Toolkit to Explore Effective On-Body Haptic Feedback by Ideating, Designing, Evaluating and Refining Haptic Designs

Using Group Collaboration. *International Journal of Human–Computer Interaction* (2024), 1–21. doi:10.1080/10447318.2024.2358460 Publisher: Taylor & Francis.

[16] Jussi Mikkonen and Christina Fyhn Nielsen. 2020. Storycoding - Programming Physical Artefacts for Research Through Design. In *Proceedings of the 2020 ACM Designing Interactive Systems Conference (DIS '20)*. Association for Computing Machinery, New York, NY, USA, 441–455. doi:10.1145/3357236.3395555

[17] Sean M. Montgomery, Nitin Nair, Phoebe Chen, and Suzanne Dikker. 2024. Validating EmotiBit, an open-source multi-modal sensor for capturing research-grade physiological signals from anywhere on the body. *Measurement: Sensors* 32 (April 2024), 101075. doi:10.1016/j.measen.2024.101075

[18] Nihar Sabnis, Dennis Wittchen, Gabriela Vega, Courtney N. Reed, and Paul Strohmeier. 2023. Tactile Symbols with Continuous and Motion-Coupled Vibration: An Exploration of using Embodied Experiences for Hermeneutic Design. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*. Association for Computing Machinery, New York, NY, USA, 1–19. doi:10.1145/3544548.3581356

[19] Laia Turmo Vidal, Yinchu Li, Martin Stojanov, Karin B Johansson, Beatrice Tylstedt, and Lina Eklund. 2023. Towards Advancing Body Maps as Research Tool for Interaction Design. In *Proceedings of the Seventeenth International Conference on Tangible, Embedded, and Embodied Interaction (TEI '23)*. Association for Computing Machinery, New York, NY, USA, 1–14. doi:10.1145/3569009.3573838

[20] Anna Vallgårda. 2014. Giving form to computational things: developing a practice of interaction design. *Personal and Ubiquitous Computing* 18, 3 (March 2014), 577–592. doi:10.1007/s00779-013-0685-8

[21] Anna Vallgårda, Laurens Boer, Vasiliki Tsaknaki, and Dag Svanæs. 2016. Material Programming: a Design Practice for Computational Composites. In *Proceedings of the 9th Nordic Conference on Human-Computer Interaction (NordiCHI '16)*. Association for Computing Machinery, New York, NY, USA, 1–10. doi:10.1145/2971485.2971554

[22] Anke van Oosterhout, Miguel Bruns, and Eve Hoggan. 2023. Designing for Haptic and Embodied Interaction with Feelix. In *Proceedings of the Seventeenth International Conference on Tangible, Embedded, and Embodied Interaction (TEI '23)*. Association for Computing Machinery, New York, NY, USA, 1–4. doi:10.1145/3569009.3571842

[23] Charles Windlin. 2023. *Shape and Being Shaped : Sketching with Haptics in Soma Design*. Ph. D. Dissertation. https://urn.kb.se/resolve?urn=urn:nbn:se:kth:diva-326780 Publisher: KTH Royal Institute of Technology.

[24] Charles Windlin, Anna Ståhl, Pedro Sanches, Vasiliki Tsaknaki, Pavel Karpashevich, Madeline Balaam, and Kristina Höök. 2019. Soma Bits - Mediating Technology to Orchestrate Bodily Experiences. *Proceedings of the 4th Biennial Research Through Design Conference* 4, Article 25 (March 2019), 1–16. doi:10.6084/

[25] Ran Zhou, Zachary Schwemler, Akshay Baweja, Harpreet Sareen, Casey Lee Hunt, and Daniel Leithinger. 2023. TactorBots: A Haptic Design Toolkit for Out-of-lab Exploration of Emotional Robotic Touch. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. ACM Press, Hamburg, Germany, 19. doi:10.1145/3544548.3580799